上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 9 and 10

Carbon Accounting: Product Carbon Footprint

Haishan Yu

饮 水 思 源 · 爱 国 荣 校



Learning Outcomes

1. **Explain** what a product carbon footprint represents and how it differs from an organizational GHG inventory.
2. **Identify** the functional unit, life-cycle boundary, activity data and emission factors used in a simplified PCF.
3. **Analyze** a product carbon footprint (PCF) results to identify a hotspot and propose a possible reduction opportunity.
4. **Use** evidence from the BMW case to evaluate a reduction option and state the limits of the conclusion.

Lecture 9

目 录

Contents

01- What PCF Is and Why It Matters

Walker’s example

02- Main Standards and How It Is Done

PAS 2050, GHGP Product Standard

Lecture 10

目 录

Contents

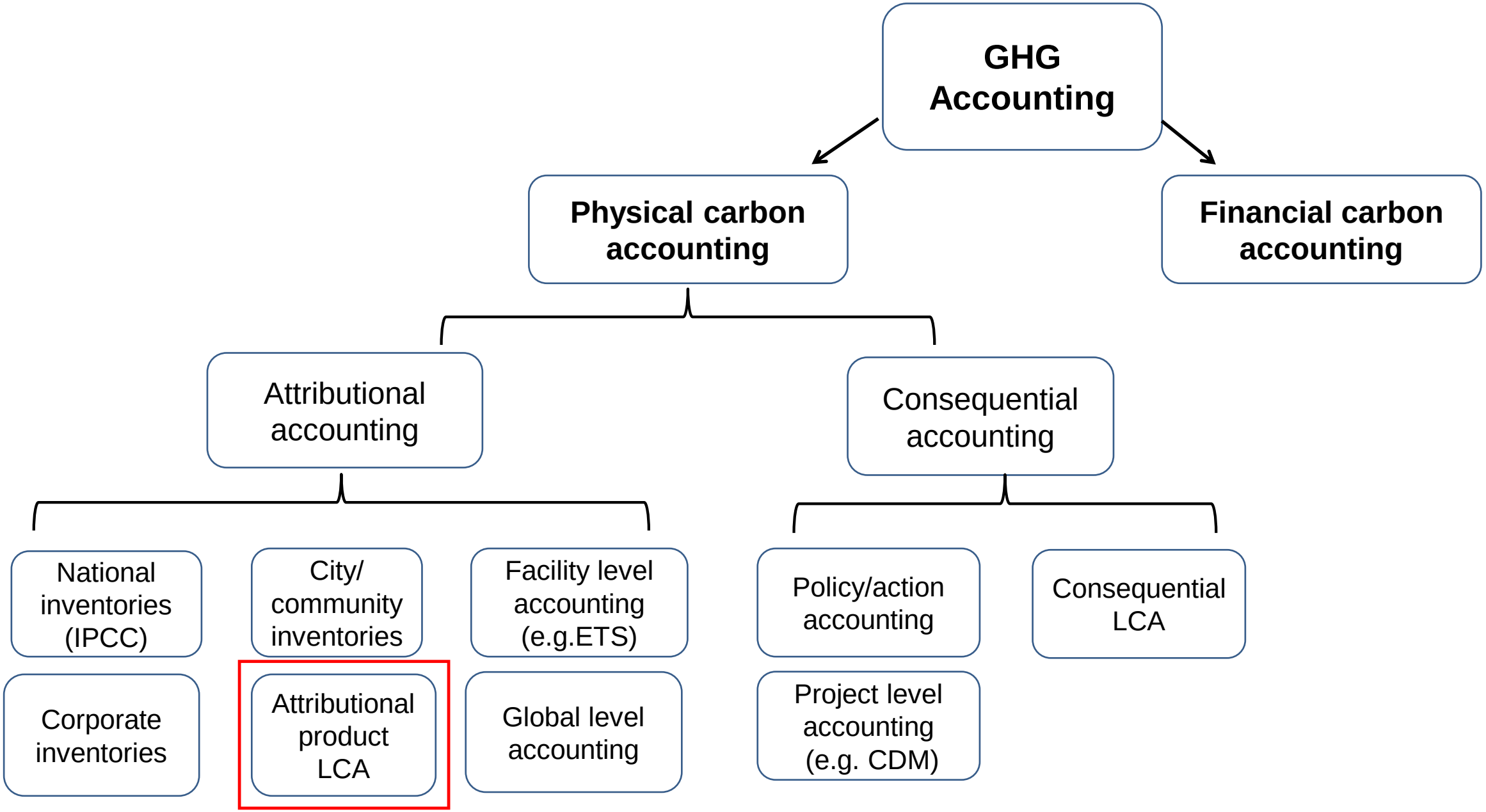
01- BMW’s Product Carbon Footprint

TBA

02- Quiz, Wrap up and Next time

Two Accounting Lenses: Corporate vs Product

	Organizational GHG Inventory	Product Carbon Footprint
Unit of analysis	the corporate	the product system
Boundary	organizational + operational	life cycle stages
Core concepts	Scope 1, 2, 3	functional unit, system boundary, allocation
Main use	disclosure, target tracking, internal management, compliance	product design, comparison, claims, compliance


GHG Accounting
Physical carbon accounting
Financial carbon accounting
Attributional accounting
Consequential accounting
National inventories (IPCC)
City/ community inventories
Facility level accounting (e.g.ETS)
Policy/action accounting
Consequential LCA
Corporate inventories
Attributional product LCA
Global level accounting
Project level accounting (e.g. CDM)

01

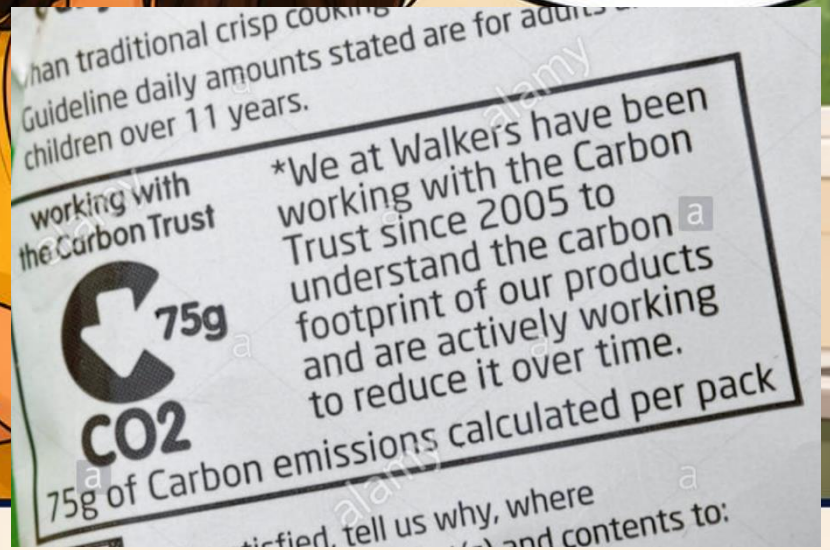
What PCF Is and Why It Matters

A Question to Start With

Mom, this packet weighs only 33.5 grams. Why does it say 75 grams of CO_2?

That's a great question. What do you think this number means?





What does 75 g CO_2 mean?

Generated by Chat GPT 5.5

投票 最多可选1项

设置

In the Walkers case, the packet weighs 33.5 grams, but the label says 75 grams CO_2. What does the 75 grams most likely mean?

A The crisps contain 75 grams of carbon dioxide.

B The packet releases 75 grams of CO_2 when opened or eaten.

C The product's life-cycle carbon emissions are estimated as 75 grams.

D Walkers' factory directly emits 75 grams CO_2 for each packet.

提交

投票 最多可选1项

设置

Before seeing any actual data — just based on your intuition —**which part of the crisps' life cycle would you expect to contribute the most** to that 75 grams?

A Raw materials — potatoes, sunflower oil, seasoning.

B Manufacturing — slicing, frying, drying.

C Packaging — plastic film and cardboard.

D Distribution — transporting crisps to stores.

E I cannot tell.

提交

How Should We Read "75 g CO_2"?

75 g CO_2

Product Carbon Footprint

Unit

gram CO_2

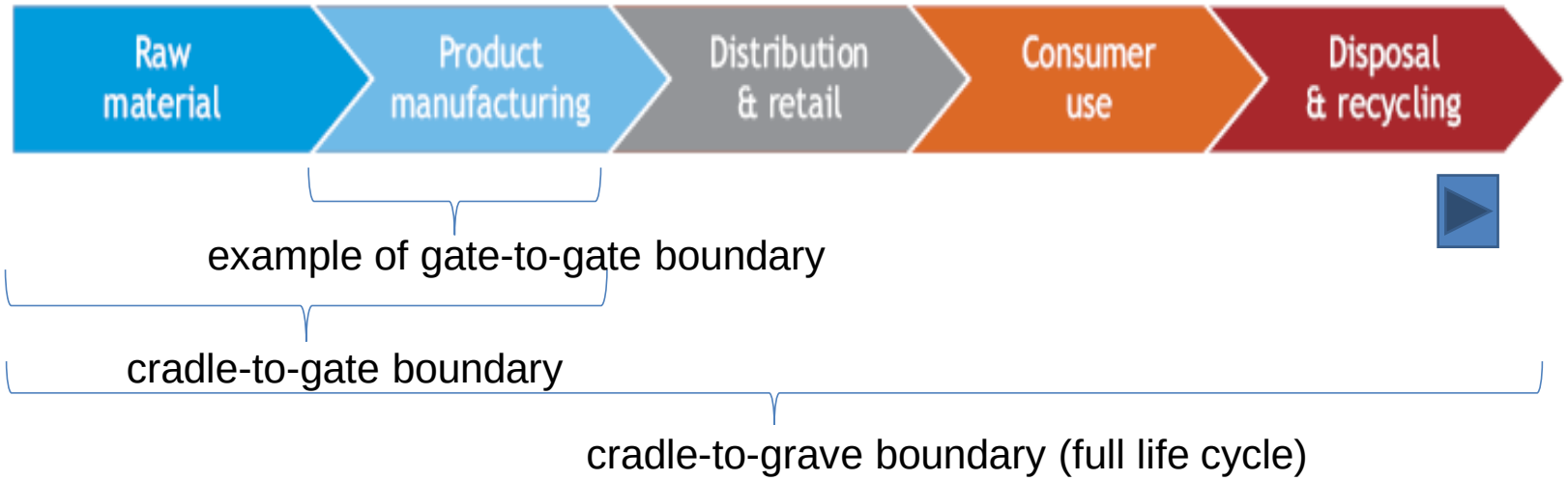
CO_2e

All greenhouse gases converted to CO_2 equivalents using GWP_{100} (IPCC)

Calculation

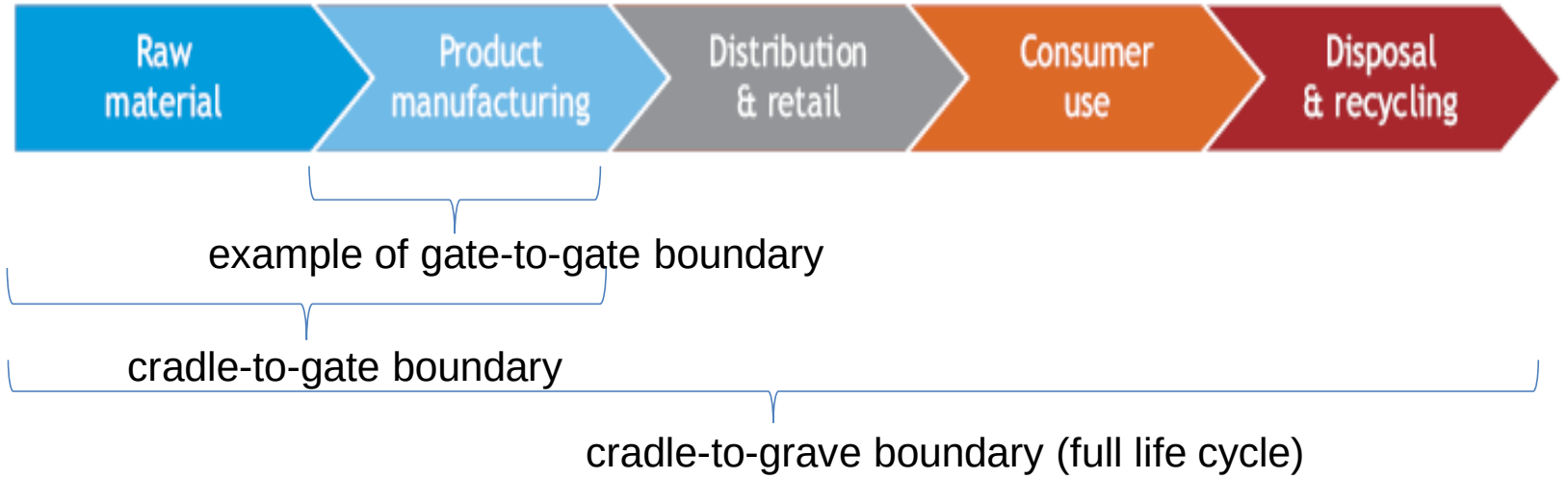
Activity data × Emission factor

Boundary



What is a Product Carbon Footprint?

- A **product carbon footprint** quantifies **greenhouse gas emissions** associated with a product [**good or service**] over a defined life cycle.





- Aluminium Production
- Sugar crop

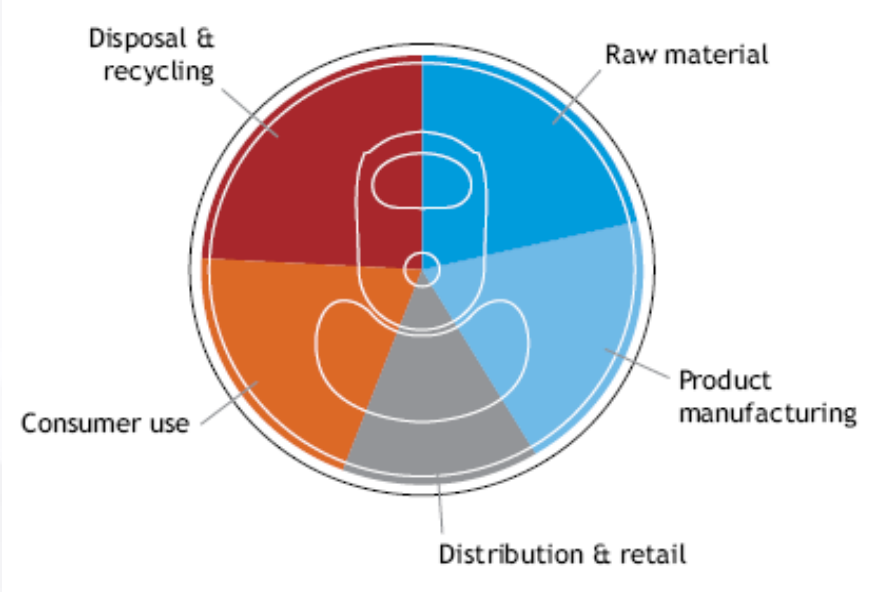
- drink production
- Packaging

- Transport
- Chilled storage and retail

- Refrigeration

- Can collection
- Recycling or disposal

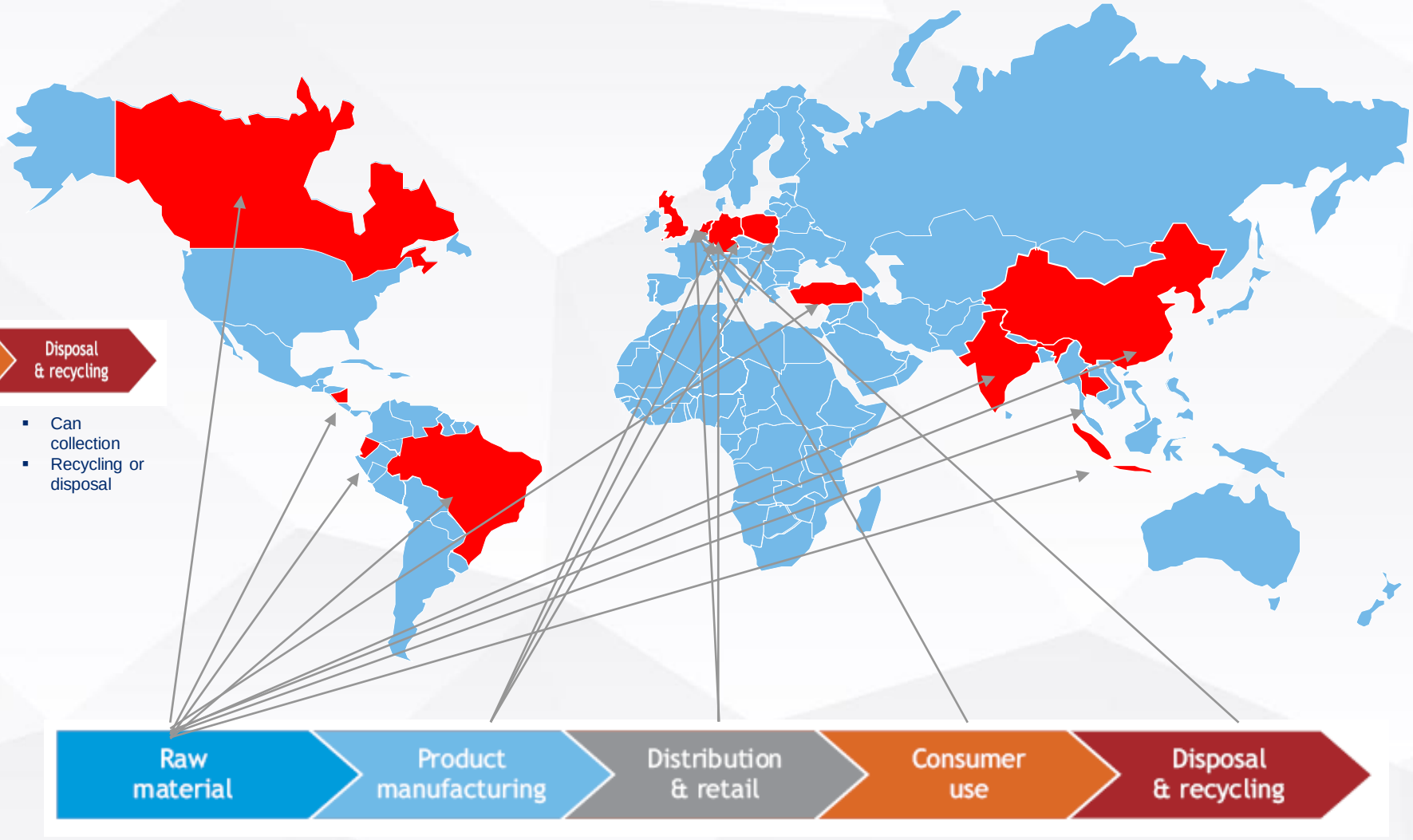
Total carbon footprint of the can of lemonade (illustrative example)



Images from the Carbon Trust



- Aluminium Production
- Sugar crop
- drink production
- Packaging
- Transport
- Chilled storage and retail
- Refrigeration
- Can collection
- Recycling or disposal



Images from the Carbon Trust

What Should Be Counted?

Cradle (farm) → Gate (factory) → Grave (disposal)

cradle-to-grave boundary

Life-cycle stages for Walkers Crisps

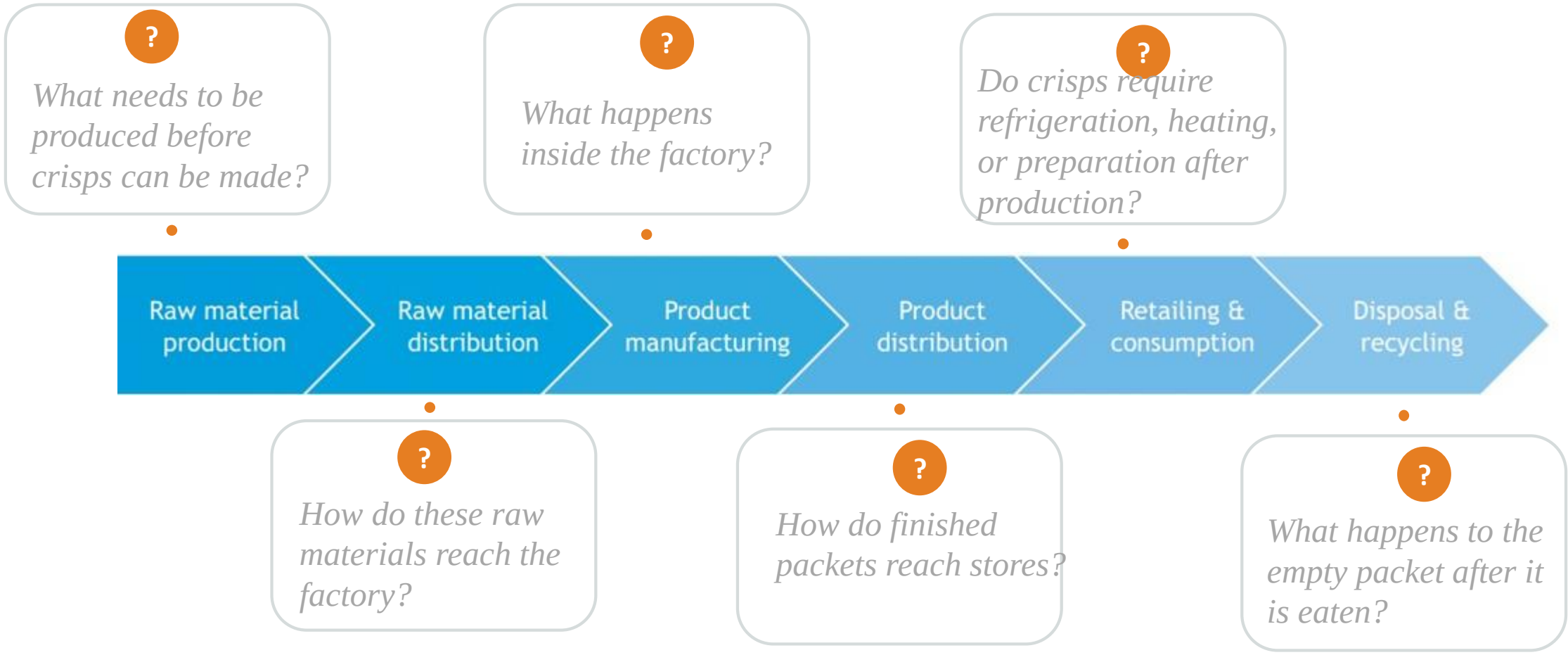
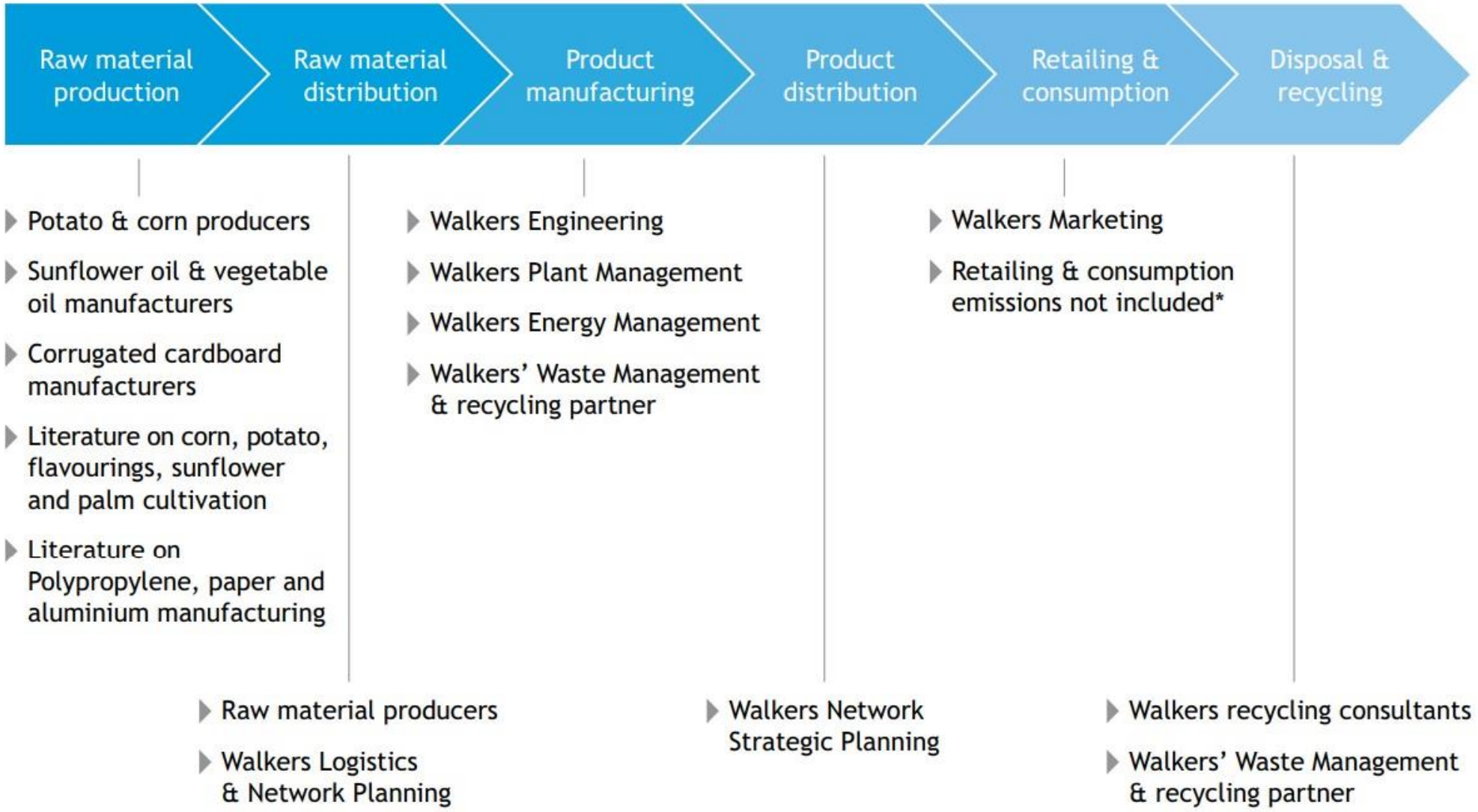


Figure 7: Companies engaged in the data collection process



*Snack foods are not refrigerated or heated in store or in the home and so retail and consumption emissions were not included.

Source: The Carbon Trust (2006), *Carbon Footprints in the Supply Chain: The Next Step for Business*.

Where Did the 75 g CO_2 Come From?

- Walkers Crisps product carbon footprint (PCF) by life-cycle stage

Before seeing any actual data — just based on your intuition —**which part of the crisps' life cycle would you expect to contribute the most** to that 75 grams?

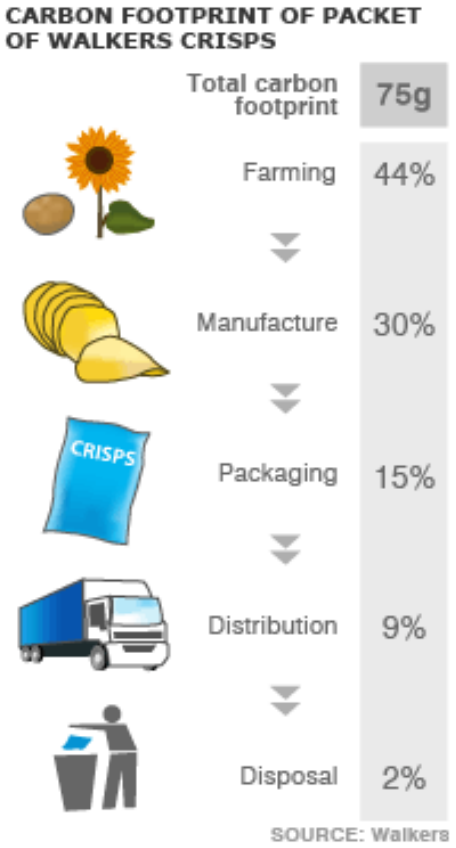
A. Raw materials — potatoes, sunflower oil, seasoning.

B. Manufacturing — slicing, frying, drying.

C. Packaging — plastic film and cardboard.

D. Distribution — transporting crisps to stores.

E. I cannot tell.



See: http://news.bbc.co.uk/2/hi/uk_news/magazine/7002450.stm

主观题 10分

设置

What are the business reasons for undertaking product carbon footprinting? (Drivers)

正常使用主观题需2.0以上版本雨课堂

作答

Why do companies conduct product carbon footprinting?

- Values and organizational alignment
  - Organizational strategy and ethics: e.g. Patagonia
  - Employee satisfaction
  - Internal communication
- Market demand and external communication
  - Customer requirements and external communication: e.g., Tesco, Walmart, CDP
  - Brand differentiation: e.g., SodaStream
  - Reputation risk
- Legislation and compliance
  - e.g., CBAM; Carbon footprint rules in EU Battery Regulation
- Resource Efficiency
  - Identify resource and cost saving opportunities: e.g., Walker's crisps
  - Managing resource risk

Product carbon footprinting (PCF) is necessarily involved with supply chain carbon accounting.

A Small Change, A Double Win

Carbon Trust helped Walkers identify a supply-chain opportunity:

a single supply-chain change could cut emissions and reduce costs.

9,200

tons CO_2/year
emissions saved

£1.2 M

per year
cost saved

Question:

Where in the supply chain might this change happen, and why?

Where in the supply chain might this change happen, and why?

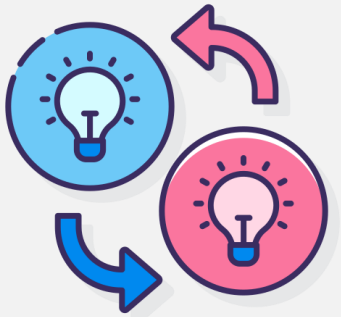
Think

30 sec

Pair

1分钟





Share

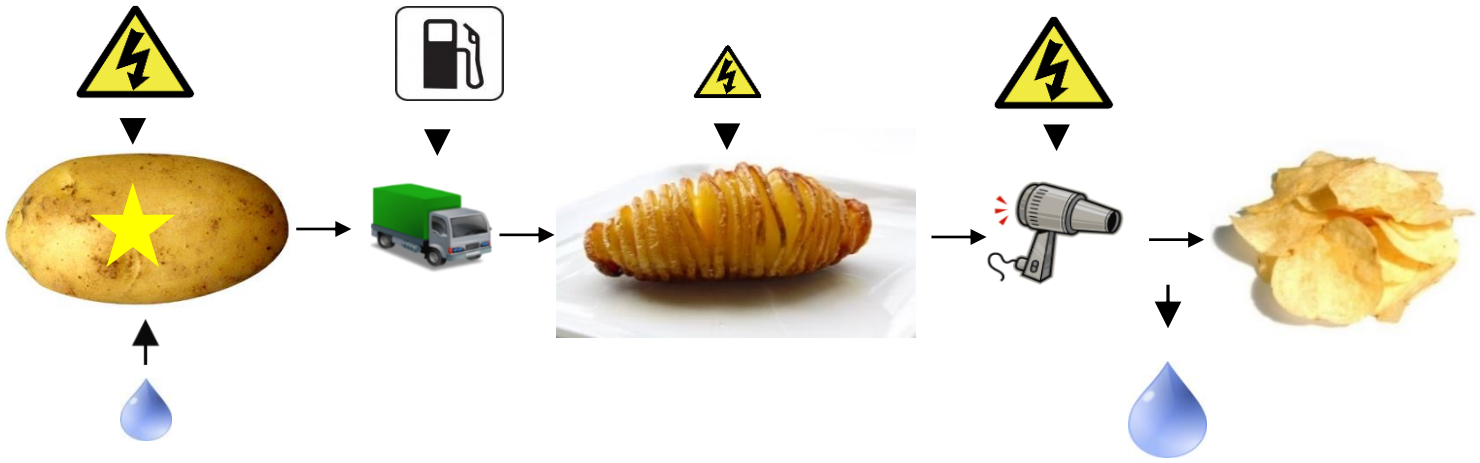
1分钟

A. **Product distribution**

B. **Packaging materials**

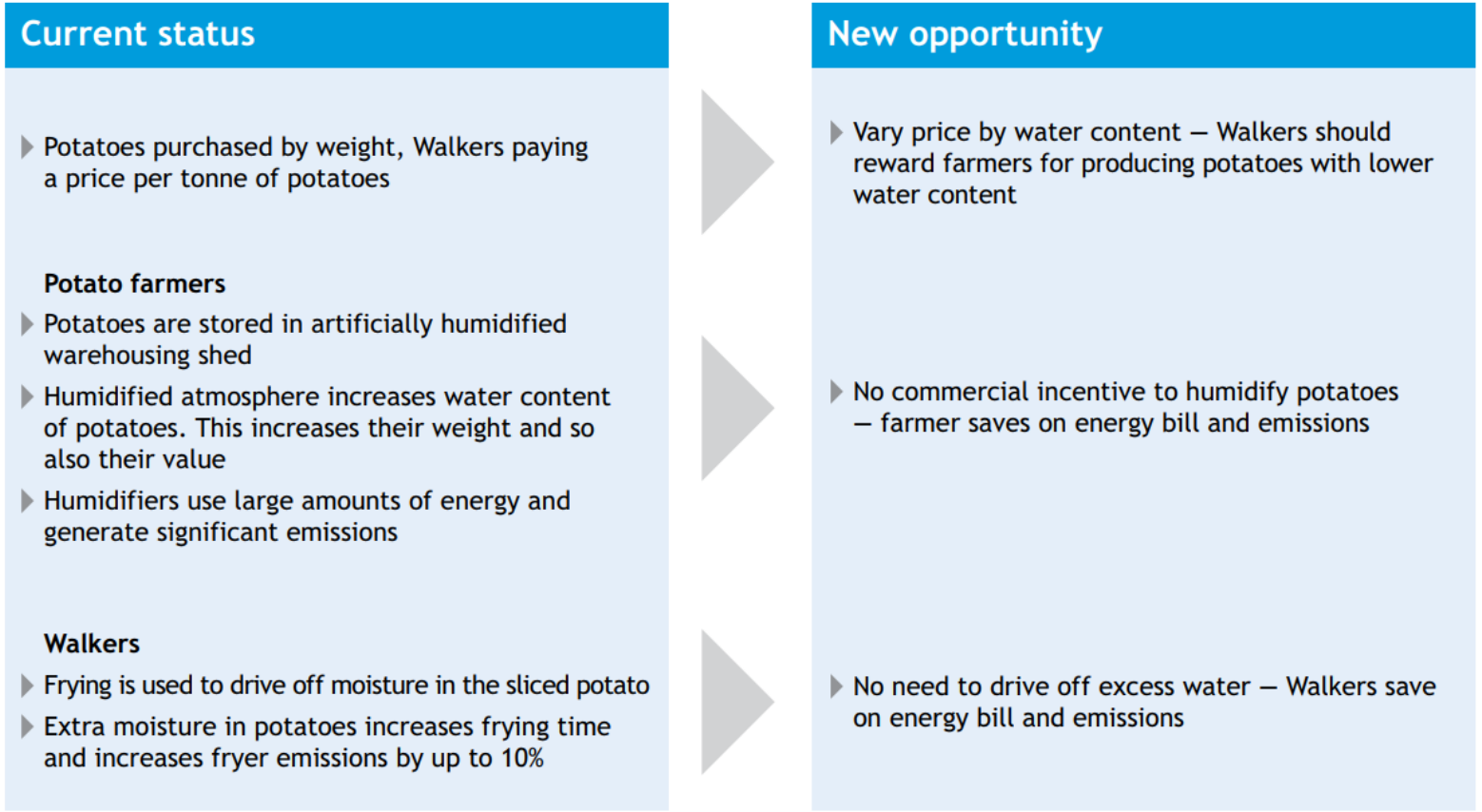
C. **Potato sourcing and purchasing**

D. **Frying and processing technology**



Walker's example

Figure 8: *Opportunity to reduce water content of potatoes*



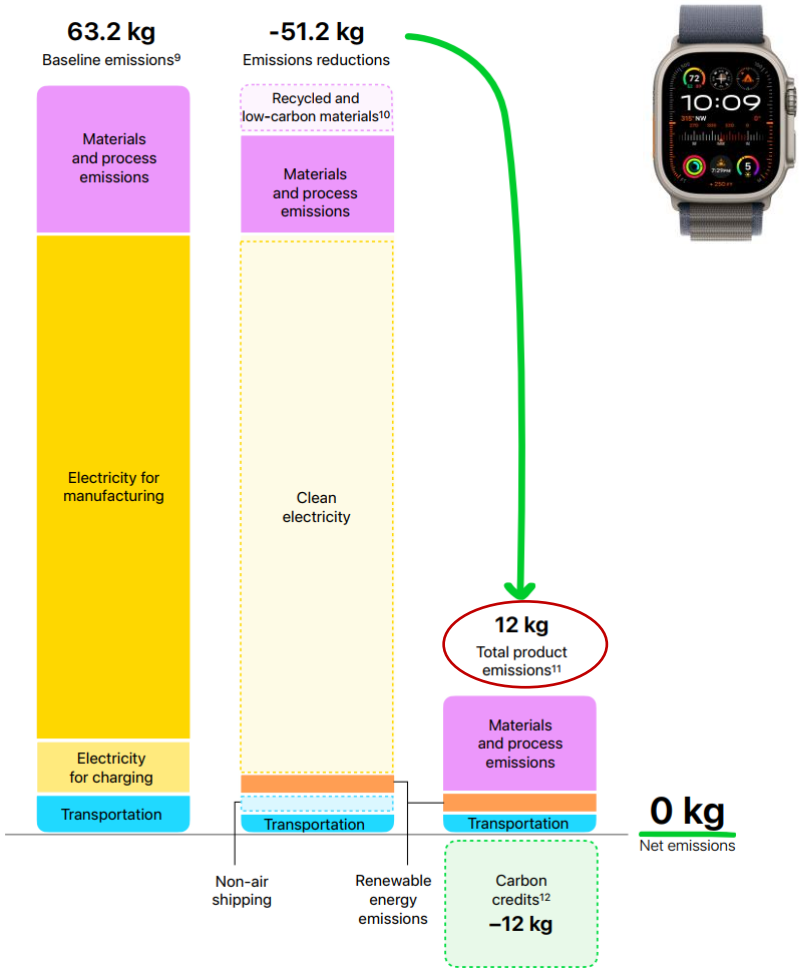
See: The Carbon Trust. (2006). *Carbon footprints in the supply chain: The next step for business* (CTC616). The Carbon Trust.

turn a division of PepsiCo. In 2007, Walkers became the first company in the world to display a carbon reduction logo on their product, and discovered considerable savings in the process. From The Economist:

"The process of calculating the carbon footprint for Walkers crisps revealed an unexpected opportunity to save energy. It turned out that because Walkers was buying its potatoes by gross weight, farmers were keeping their potatoes in humidified sheds to increase the water content. Walkers then had to fry the sliced potatoes for longer to drive out the extra moisture. By switching to buying potatoes by dry weight, Walkers could reduce frying time by 10% and farmers could avoid the cost of humidification. Both measures saved money and energy and reduced the carbon footprint of the final product."

Check Your Understanding: Apple Watch

- Now let's apply the same thinking to a different product: Apple Watch Ultra 2.



Q1: Before seeing Apple's result, which activities should be considered in Apple Watch 's PCF?
Select all that apply.

A. Extraction and production of materials
B. Manufacturing and assembly
C. Product transport and distribution
D. Electricity used during product use
E. End-of-life treatment or recycling
F. Carbon credits used to support a carbon-neutral claim

Q2: If you worked in Apple's sustainability team, where would you start reducing emissions?

A. Materials B. Manufacturing and assembly
C. Product transport D. Product use
E. End-of-life treatment

02

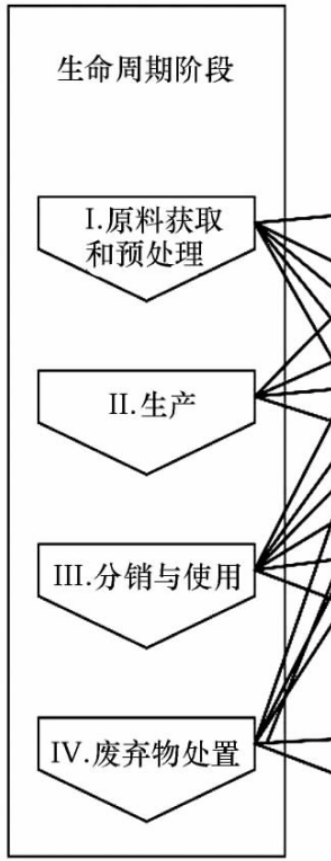
Main Standards and How it is done

Product Carbon Footprinting Standards

- A few existing standards and guidance for product carbon footprint:
  - PAS 2050
    - First published in 2008, and revised in 2011
    - the first product carbon footprinting standard
  - GHG Protocol Product standard
    - based on an life cycle and attributional approach and builds on the ISO standards for LCA
  - ISO 14067:2018
  - Others:
    - Product Environmental Footprint by EC
    - Country specific standards: e.g., BP X30-323 in France
    - EN 15804



What are the differences between these standards? (*)



产品碳足迹核算要素	核算要素的含义	产品碳足迹标准的相应规定		
		PAS2050	GHG Protocol	ISO14067
截断规则	按照重要性阈值排除不重要的贡献的规则	定量规定	无硬性定量规定	无定量规定
固定资产	产品生命周期中需要的机械、设备和基础设施的生产维护引起的排放	排除	不强制,可核算	无规定
场所运行	场所如工厂和零售点的照明、温度控制等过程的排放	核算	不强制,可核算	无规定
雇员交通	雇员通勤交通引起的排放	排除	不强制,可核算	无规定
碳存储	产品所含碳,如林产品所含的生物碳或生命周期过程中吸收的碳(如水泥使用阶段)	核算	核算	排除
消费者交通	消费者购买产品的交通过程引起的排放	排除	不强制,可核算	无规定
废物焚烧能量回收	废弃物焚烧产生的能量回收利用(如用于发电或供暖等)	排除	核算	核算
延迟排放加权系数	用于估算随时间推移延迟排放的加权平均影响的系数	不强制,可核算	排除	无规定

图 1　3 个国际产品碳足迹标准核算对核算要素的规定差异

Fig.1　The different rules on accounting elements between three international product carbon footprint standards

From: 李楠，刘盈，王震．2020．国际标准差异对产品碳足迹核算的影响分析———以胶版印刷纸为例［J］．环境科学学报，40(2) : 707-715

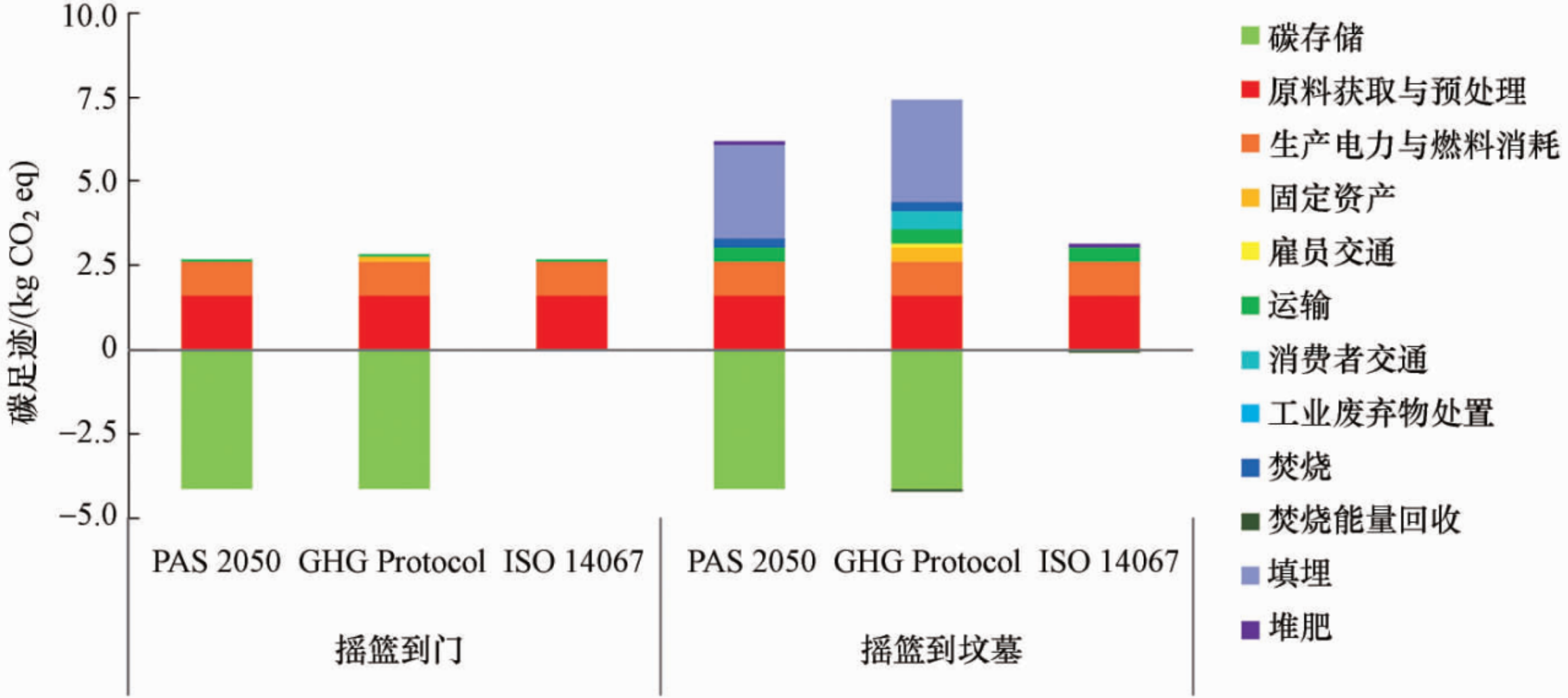
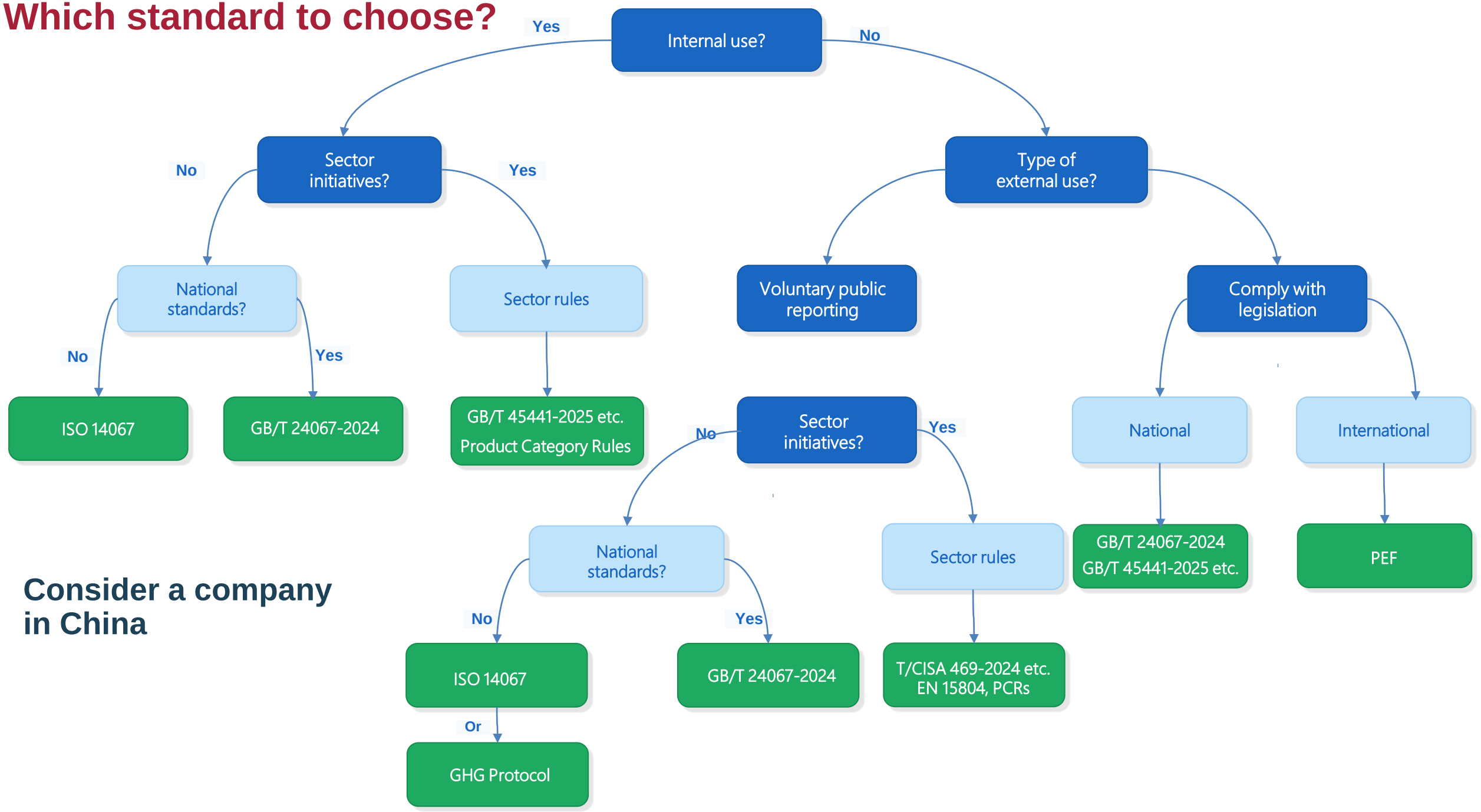


图 4　3 个标准下产品碳足迹中各排放源的贡献

Fig.4　Contributions of emission sources in product carbon footprints under three standards

From: 李楠，刘盈，王震．2020．国际标准差异对产品碳足迹核算的影响分析———以胶版印刷纸为例［J］．环境科学学报，40(2) : 707-715

Which standard to choose?



Consider a company in China

Product Carbon Footprinting measurement process

Scope and Boundary

- Determine product (or service) you are interested in footprinting.
- Determine functional unit: the performance characteristics and services delivered by the product being studied.
  - PCF results are expressed as $kgCO_2e$ per functional unit.
- Determine boundary you would like to use (Cradle-to-Gate or Cradle-to-Grave).
- Determine what gases you should take into account and GWP values.
- Develop a process map:
  - Diagrammatic representation of the life cycle and boundary.

Some examples:

Source: https://www.environdec.com/EPD-Search/

Functional unit: the quantified function provided by the product, e.g. '100g of protein'

Results are expressed 'per functional unit' e.g. '10kgCO_2/100g protein'



= 7.26 kgCO_2/product

14.22 kgCO_2/product =



Not very clear functional units

Clear functional unit



= 7.78 kgCO_2/one passenger travelling 100km

Ecolab Example from the GHG Protocol's Product Standard

Ecolab, the global leader in cleaning, sanitizing, food safety, and infection prevention products and services, performed a GHG inventory on the life cycle of their APEX™ automatic warewashing system. Ecolab selected the function as the delivery of clean and sanitized dishes through an automatic dish machine, which included the necessary individual functions that the APEX™ warewashing system provides (APEX™ Power, APEX™ Rinse Additive, and APEX™ Presoak). They chose the magnitude and duration of the function as its use in a typical food service facility for one year and set the expected level of quality as "clean and sanitized," which requires 180 °F water during use.

Using this information, the functional unit was defined as delivering clean and sanitized dishes through an automatic dish machine in a typical food service facility for one year.

The reference flow was defined as the total pounds of product required to fulfill the function, namely:

- 500 racks per day of dishes washed at a typical location with 360 operating days per year
- 1800 parts per million (ppm) average detergent concentration within the dish machine (steady-state assumption)
- 0.15 grams of rinse additive per rack of dishes
- 4000 ppm presoak concentration, dispensed twice per day

By defining a detailed functional unit – considering all functions, quality, magnitude, and duration – Ecolab was able to quickly and accurately define their reference flow. Additionally, the information collected about the use of the product was used during boundary setting (chapter 7) to easily define the use profile.

A simplified process map for a coke



Images from the Coca Cola Company

Product Carbon Footprinting measurement process

Scope and Boundary → Collect Data

- Two groups of data are necessary to calculate a carbon footprint:
  - **Activity Data:** the data which reflects the amount of material and energy involved in the product's life cycle (material inputs and outputs, energy used, transport, etc.)
  - **Emission Factors (EFs):** a conversion factor to determine the emissions from the amount of materials/energy used (e.g. kg CO_2e/kwh of electricity used).

- These data can be clarified into two broad categories in terms of quality:
  - **Primary data**: direct measurements made internally or by someone else in the supply chain (e.g. meter readings, weighed material, etc.)
  - **Secondary data**: data obtained from sources other than direct measurement of the process included in the life cycle of the product (e.g. industry report)

Primary vs. Secondary data:

- In general, use as much primary data as possible.
  - Both **PAS 2050** and the **GHGP Product Standard** require primary activity data to be used for all processes which are owned and operated by the footprinting organisation.
- Where secondary data must be used, preference shall be given to data which is:
  - Time related
  - Geographically specific
  - Technologically specific
  - Accurate
  - Precise
- In some cases secondary data may be preferable to enable consistency and, where possible, comparability:
  - For example: GWPs, Electricity EFs, disposal scenarios, etc.

Product Carbon Footprinting measurement process

Scope and Boundary | Collect Data | Calculate

- In general the following steps should be used to calculate emissions:

1. *Activity data* x *EF= emissions* / *input*, *output*, *energy* or *fugitive emissions source*. Ensure all emissions are expressed as CO_2 equivalents using GWPs.
2. Determine whether any **allocation** needs to be done to make the data product specific and ensure any co-products are properly allocated emissions.
3. Repeat the above for all sources of emissions within the agreed upon boundary.

Check if there is any co-product

Figure [9.1] Illustrative generic common process that requires allocation[1]

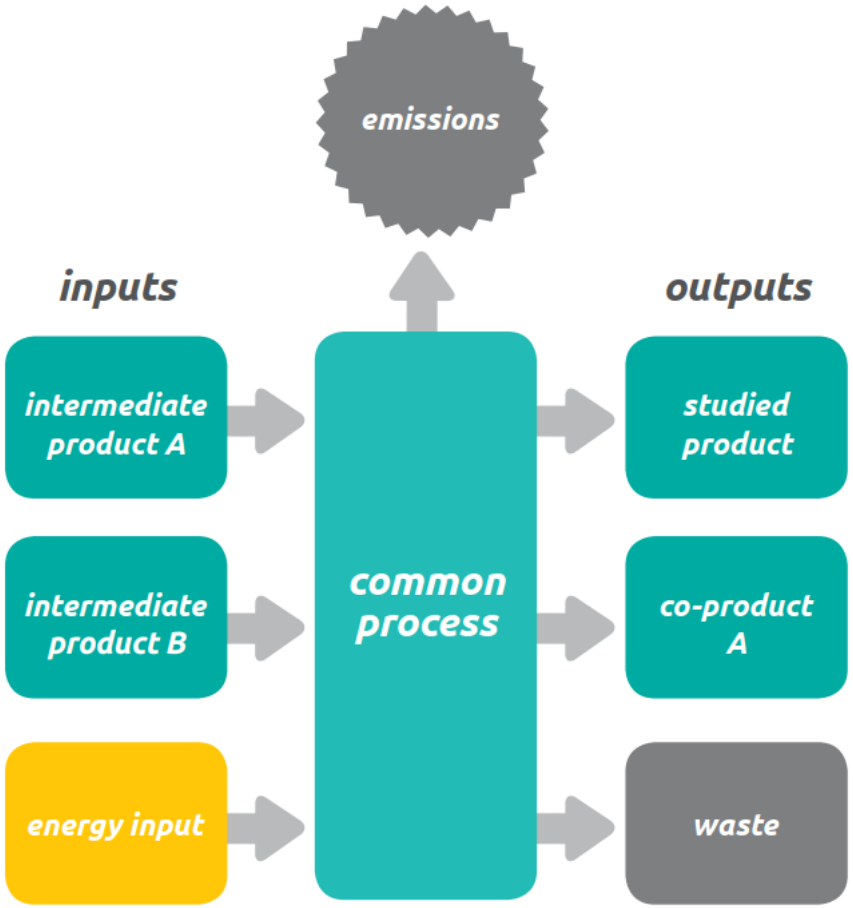


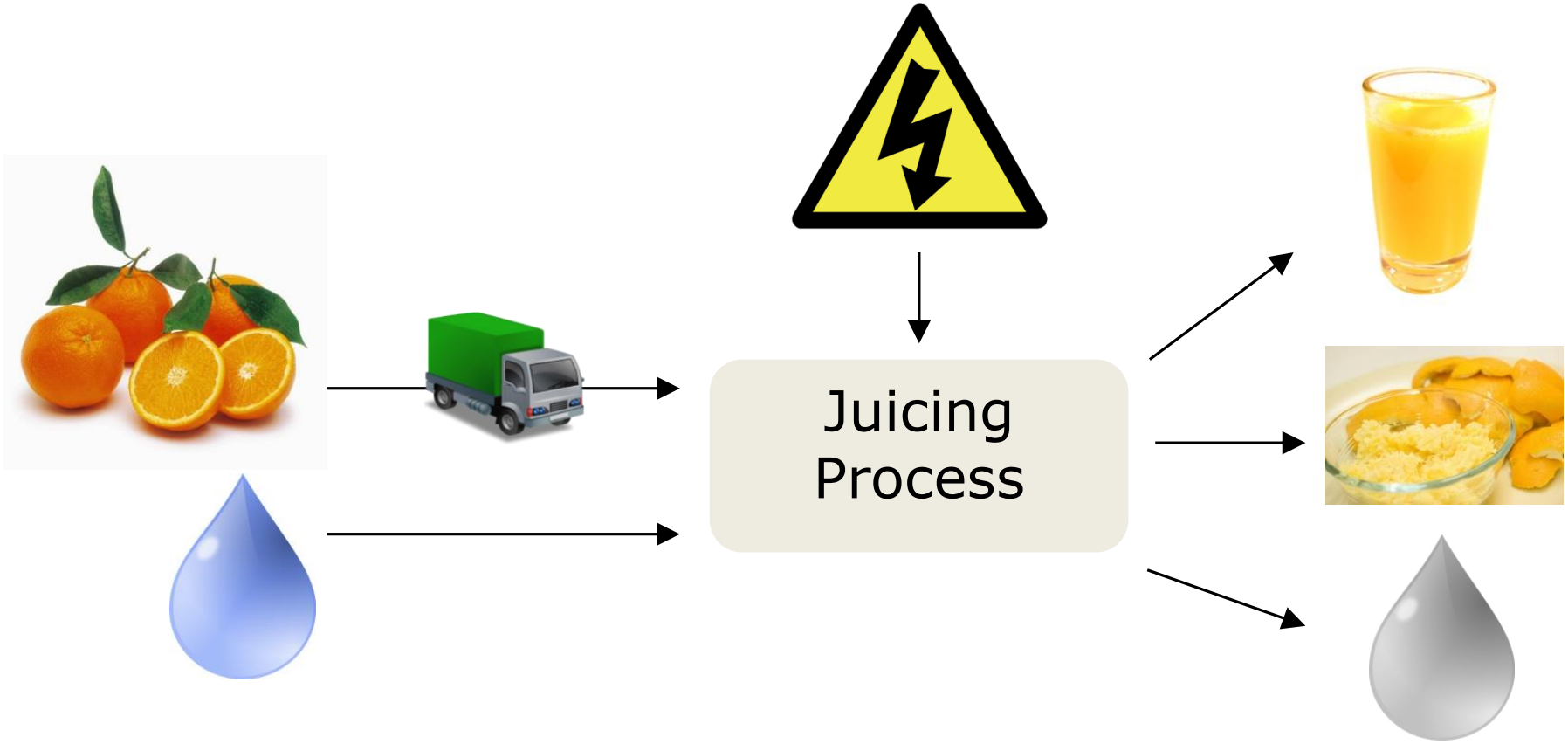
Table [9.1] Methods to avoid allocation

Method	Definition
Process subdivision	Dividing the common process into sub-processes.
Redefining the unit of analysis	Inclusion of the co-products (additional functions) in the functional unit.
System expansion	Using the emissions from an alternative product that comprises the same functional unit as a co-product to estimate the emissions of the co-product. Only applicable when companies have direct knowledge of the function and eventual use of the co-product.

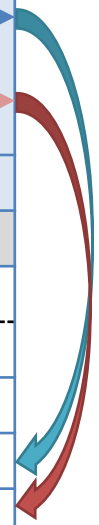
Table [9.2] Method to perform allocation

Method	Definition
Physical allocation	Allocating the inputs and emissions of the system based on an underlying physical relationship between the quantity of product and co-product and the quantity of emissions generated.
Economic allocation	Allocating the inputs and emissions to the product and co-product(s) based on the market value of each when they exit the common process.
Other relationships	Allocating the inputs and emissions to the product and co-product(s) based on established and justifiable relationships other than physical or economic.

Example of allocation:



Process	Emissions source		Activity Data	Unit	Emissions Factor	Unit	Allocation	Emissions ($kgCO_2e$) AD*EF*Alloc
Orange Pressing (per 1000L of Juice output)	Brazilian Oranges	Input	1842	kg	0.1245	$kgCO_2e/kg$		229.33
	Clean Water		2	L	0.0003	$kgCO_2e/L$		0.0006
	Orange Juice	Output/ Co-Product	1000	kg	0	$kgCO_2e/kg$	97%	
	Peel/Pith		745	kg	0	$kgCO_2e/kg$	1%	
	Pulp		98	kg	0	$kgCO_2e/kg$	2%	
	Waste Water		1	L	0.0005	$kgCO_2e/L$		0.0005
		Balances?	YES					
	Oranges	Transport	1842	kg	0.541	$kgCO_2e/kg$		996.52
	Electricity (Brazil)	Energy	66	kwh	0.1212	$kgCO_2e/kwh$		8.00
						Total from Process ($kgCO_{2e}$/1000L)	100%	1233.85
						Total allocated to juice ($kgCO_{2e}$/1000L)	97%	1196.84
						Total allocated to co-products ($kgCO_{2e}$/1000L)	3%	37.02



To better understand this result, what would you like to examine most closely in this calculation, and why?

Some Questions Behind the Calculation

Orange juice case

1,196.84

kg CO_2e per 1,000 L of juice

Calculated from activity data, emission factors
and a 97% allocation to orange juice

- **Which life-cycle stages do we include, and does this boundary fit our purpose?**

  The boundary should reflect the accounting purpose. We need to justify exclusions and consider how they affect the result.

- **Can we obtain data that represent this product?**

  Supplier data may be unavailable or confidential. Available datasets may describe a different technology, location or year.

- **How do our modelling choices affect the result?**

  Allocation rules, estimates and assumptions shape the calculation.

- **Can someone else trace and check the calculation?**

  Checking the result requires clear records of data sources, methods and assumptions, often gathered from many value-chain partners.

The calculation gives a result. Its credibility depends on the data and choices behind it.

Wrap up – Product Carbon Footprint

Why – why firms conduct PCF?

Internal

Identify resource and cost saving opportunities

External

What – what is counted?

A defined boundary (life cycle)

Functional unit

Gases, Reporting year

How – how is it quantified?

Activity Data * Emission Factors

Primary data

Secondary data

✓ **Key Concepts**

1. A product carbon footprint (PCF) quantifies life-cycle GHG emissions within a defined boundary — it is NOT the physical weight of the product.

2. Three key elements: Unit (CO_2e) | Boundary | Calculation

3. Life-cycle stages: from raw materials to manufacturing, distribution, use and disposal & recycling.

Product Environmental Report

Apple Watch Ultra 2 — Carbon Neutral



After-class Question:

Why was Apple's carbon-neutral Apple Watch claim questioned?





See you next time

Lecture 8

目 录

Contents

01- BMW’s Product Carbon Footprint

02- Quiz, Wrap up and Next time

03

BMW and PCF

If a BEV always the lower-carbon option?

A BEV has zero tailpipe emissions.

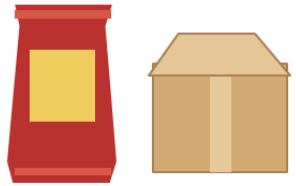
Does it always have a lower life-cycle carbon footprint than an ICE vehicle?

A simplified process map for a coke



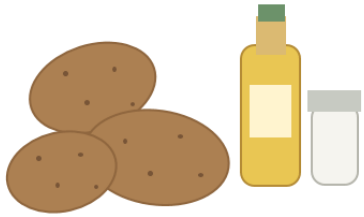
Images from the Coca Cola Company

Walkers Crisps: Product Process Map



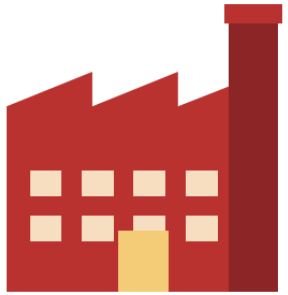
Packaging

Bags and cartons



Ingredients

Potatoes, oil and seasoning



Manufacturing

Wash, peel, slice, fry, season and pack



Potato peels



Other uses

Further processing



Distribution

Storage and transport



Retail

Shops and supermarkets



Consumption

Eating the crisps



Recycling

Accepted packaging materials



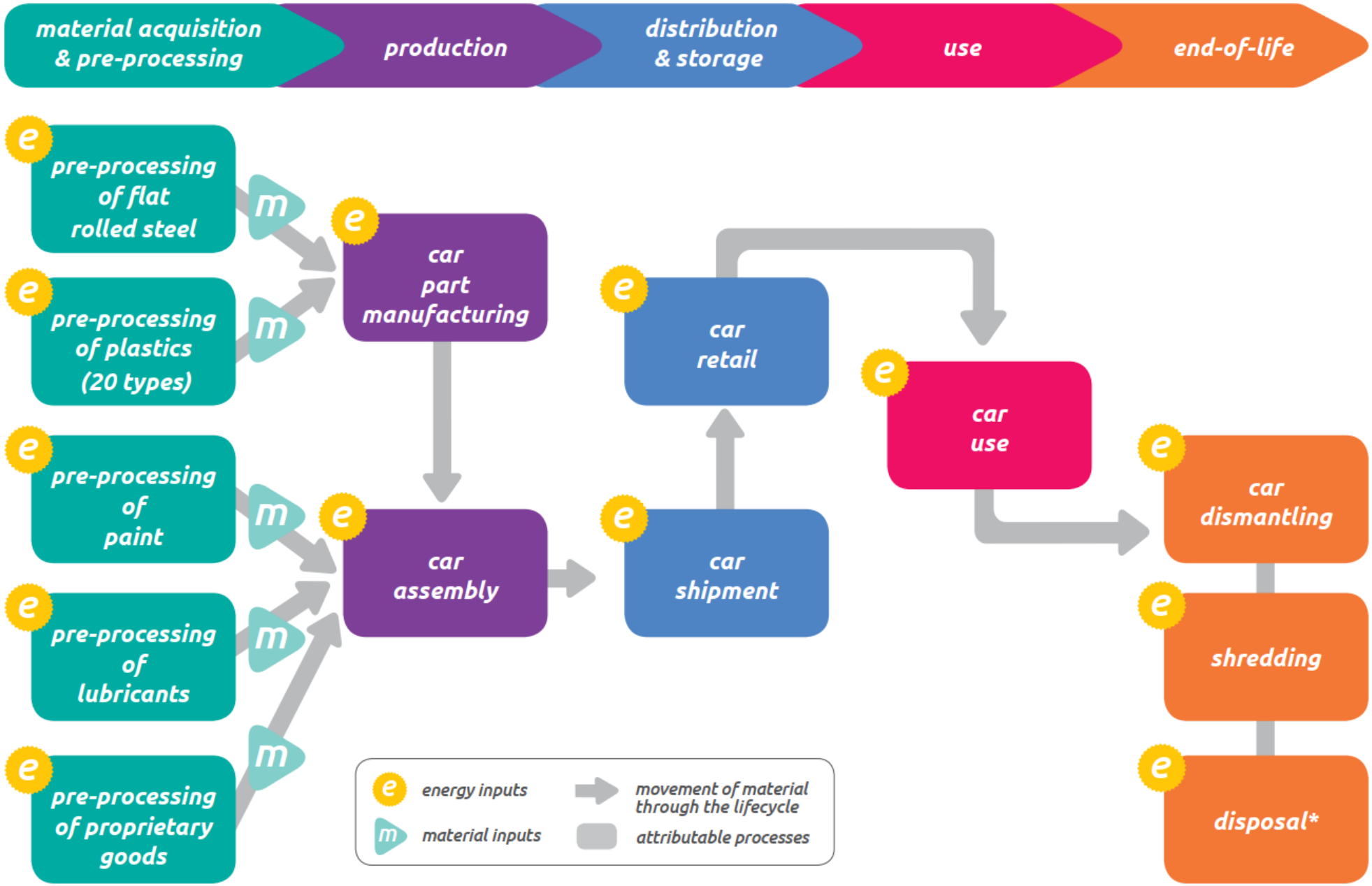
Residual waste

Incineration / landfill

Group Activity

Draw a process map for a car. Show the main stages and use arrows to connect them.

Figure [7.2] Sample process map for a car (cradle-to-grave inventory)



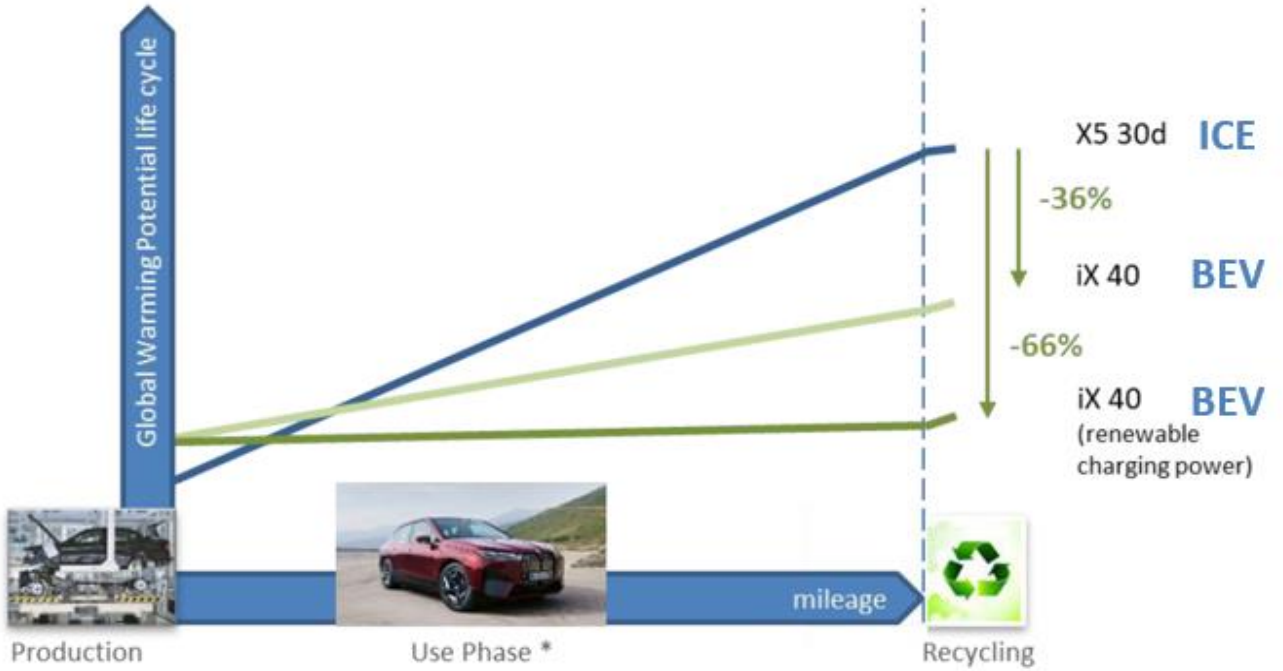
* Recycling of parts is not included in this simplified example

BMW Vehicle Life-cycle Hotspots

Life-cycle stage	ICE vehicle	BEV
Materials and components	Steel, aluminum, plastics	Steel, aluminum, plastics, battery materials
Battery	Not central	**Major upstream hotspot**
Manufacturing	Vehicle production	Vehicle and battery-related production
Use phase	**Fuel combustion**	Charging electricity
End-of-life	Recycling / disposal	Battery recovery and material circularity

BEVs reduce use-phase tailpipe emissions but increase the importance of upstream emissions.

Three Assumptions That Shape the Result



Battery production

- the emissions right out of production

Life cycle distance driven

- the length (mileage) of the chart

Grid emissions

- the slope of the lines

Source: "Environmental Report BMW iX xDrive40," BMW Life Cycle Assessment, Figure 3, page 4, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/responsibility/downloads/en/2021/Environmental-report_BMW-iX.pdf, accessed April 2024.

Vehicle Lifecycle Emissions Calculator (Harvard Business School Case 123-008)

BMW’s x5 and ix5 mode’s PCF

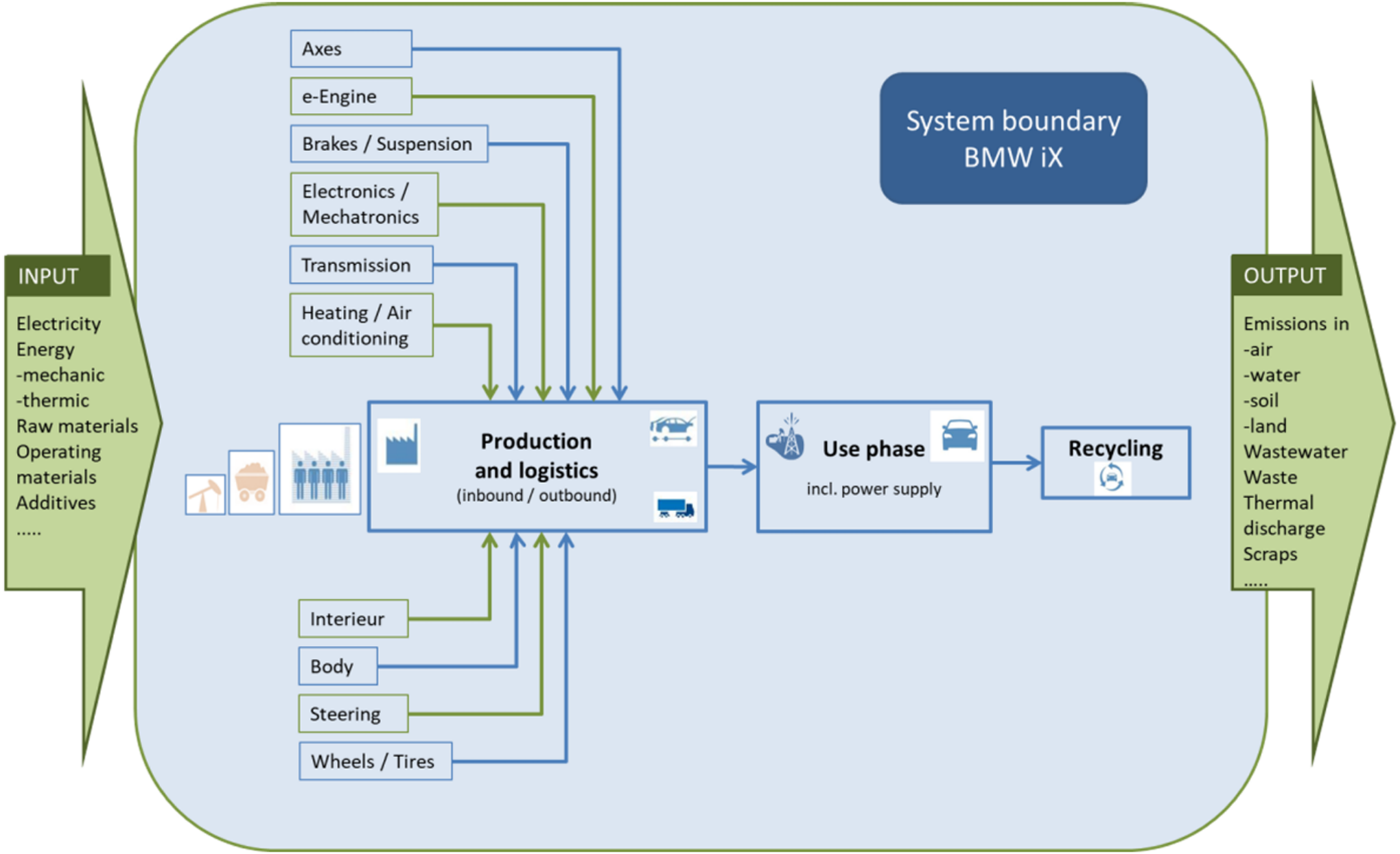


Fig. 1: Flowchart input / output data of the BMW iX

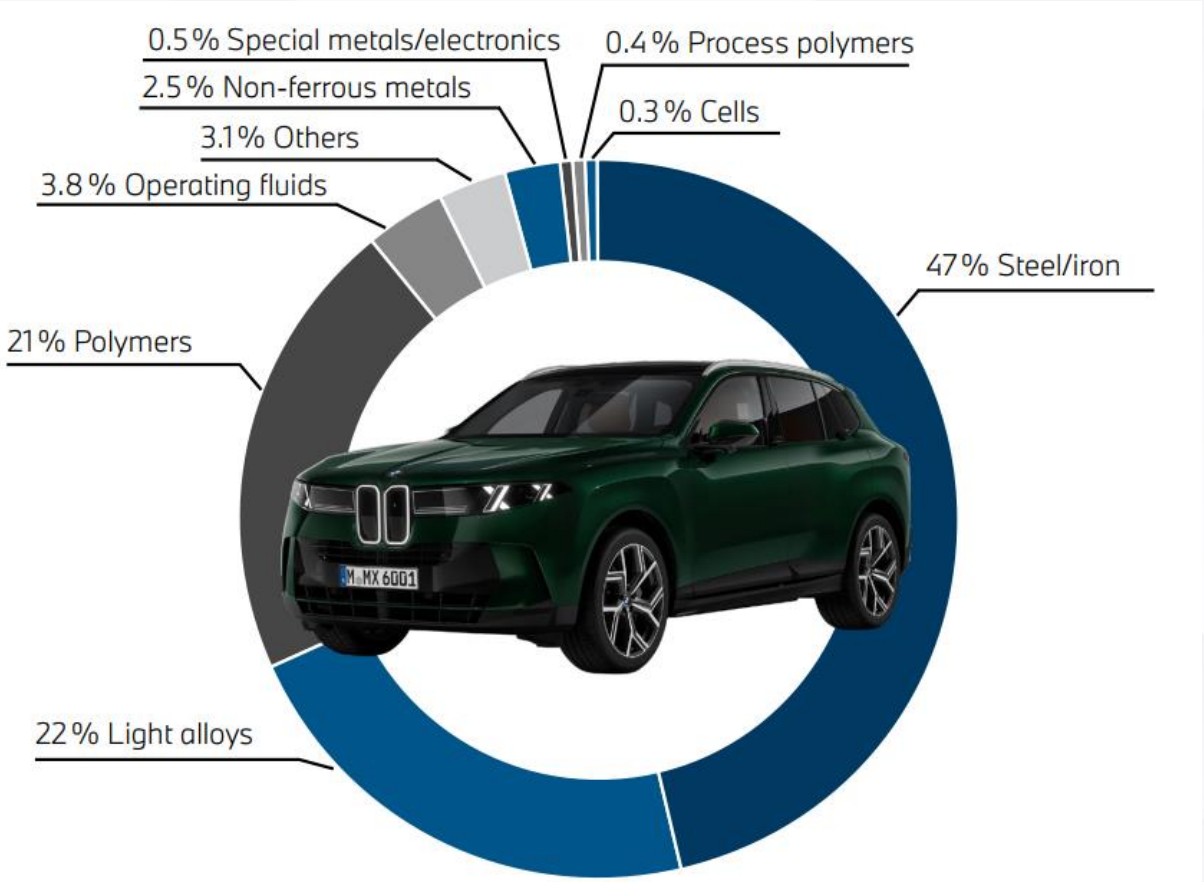


Fig. 2: Material composition of the BMW X5 40d xDrive at the start of production.
The specified values may contain rounding differences.

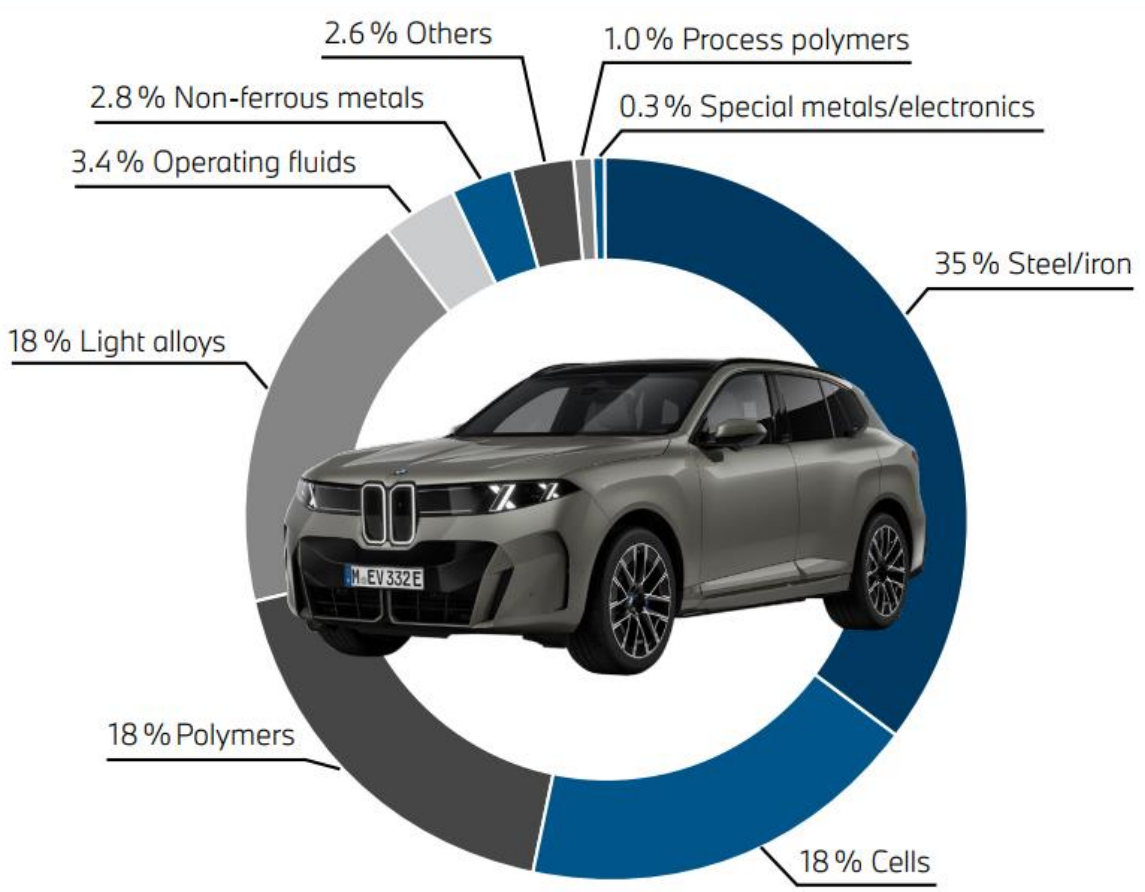


Fig. 2: Material composition of the BMW iX5 60 xDrive at the start of production.
The specified values may contain rounding differences.

Carbon Footprint [CO_2e] of the BMW X5 40d xDrive over its life cycle

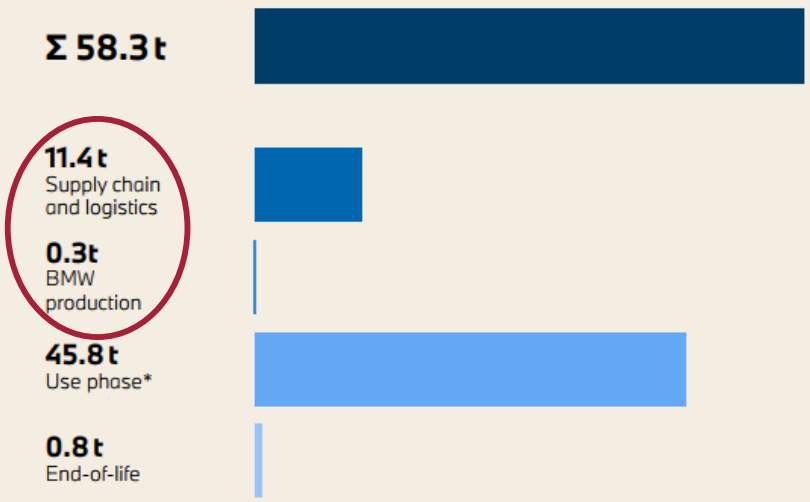


Fig. 3: The total amount of carbon dioxide (CO_2) and other greenhouse gas emissions such as methane or nitrogen oxide are taken into account. CO_2 equivalents (CO_2e) are a unit of measurement used to standardise the climate impact of different greenhouse gases. The crediting of electricity from renewable energy sources includes both electricity from renewable in-house-generation plants and direct supply contracts as well as certified guarantees of origin. Offsetting measures are not taken into account.

* Consumption data based on WLTP best-case value plus a segment-specific average mark-up for optional equipment in accordance with registrations in 2025 in EU27 (+UK, CH).

Carbon Footprint [CO_2e] of the BMW iX5 60 xDrive over its life cycle

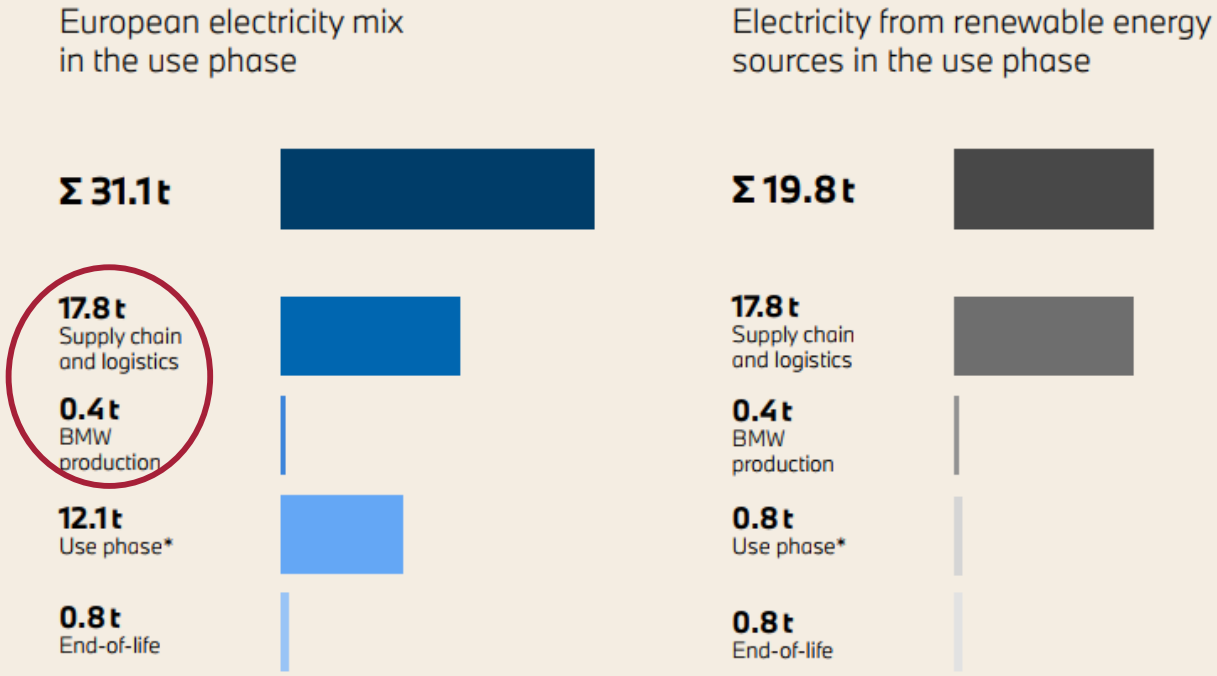


Fig. 3: The total amount of carbon dioxide (CO_2) and other greenhouse gas emissions such as methane or nitrogen oxide are taken into account. CO_2 equivalents (CO_2e) are a unit of measurement used to standardise the climate impact of different greenhouse gases. The crediting of electricity from renewable energy sources includes both electricity from renewable in-house-generation plants and direct supply contracts as well as certified guarantees of origin. Offsetting measures are not taken into account.

* Consumption data based on WLTP best-case value plus a segment-specific average mark-up for optional equipment in accordance with registrations in 2025 in EU27 (+UK, CH).

ICE **EV**

ICE

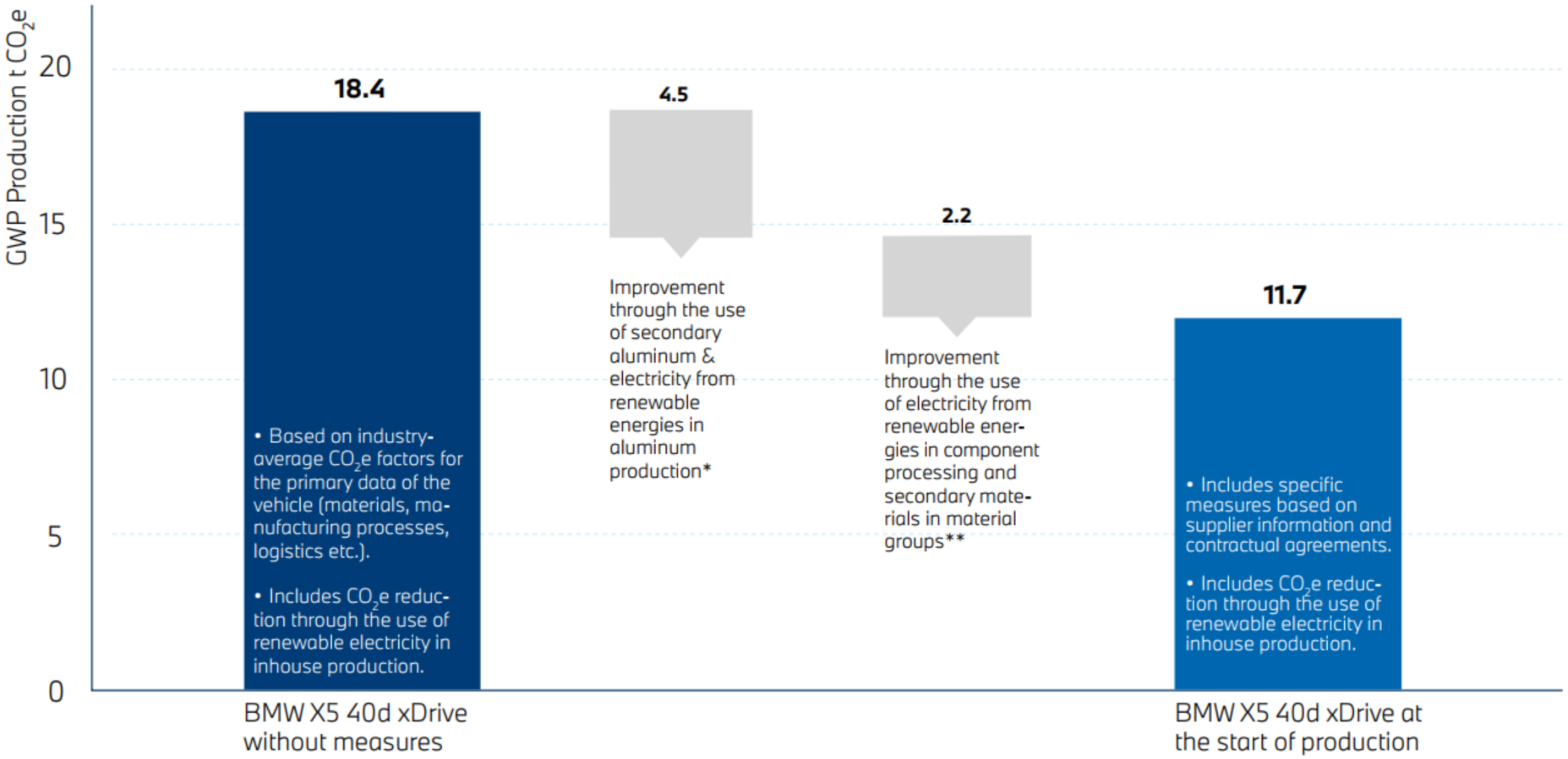


Fig. 5: Influence of development targets on the Carbon Footprint in the manufacturing phase of the BMW X5 40d xDrive.
* Wheels, body and drivetrain components.
** Secondary material content in the following materials: steel, plastics, copper, glazing.

EV

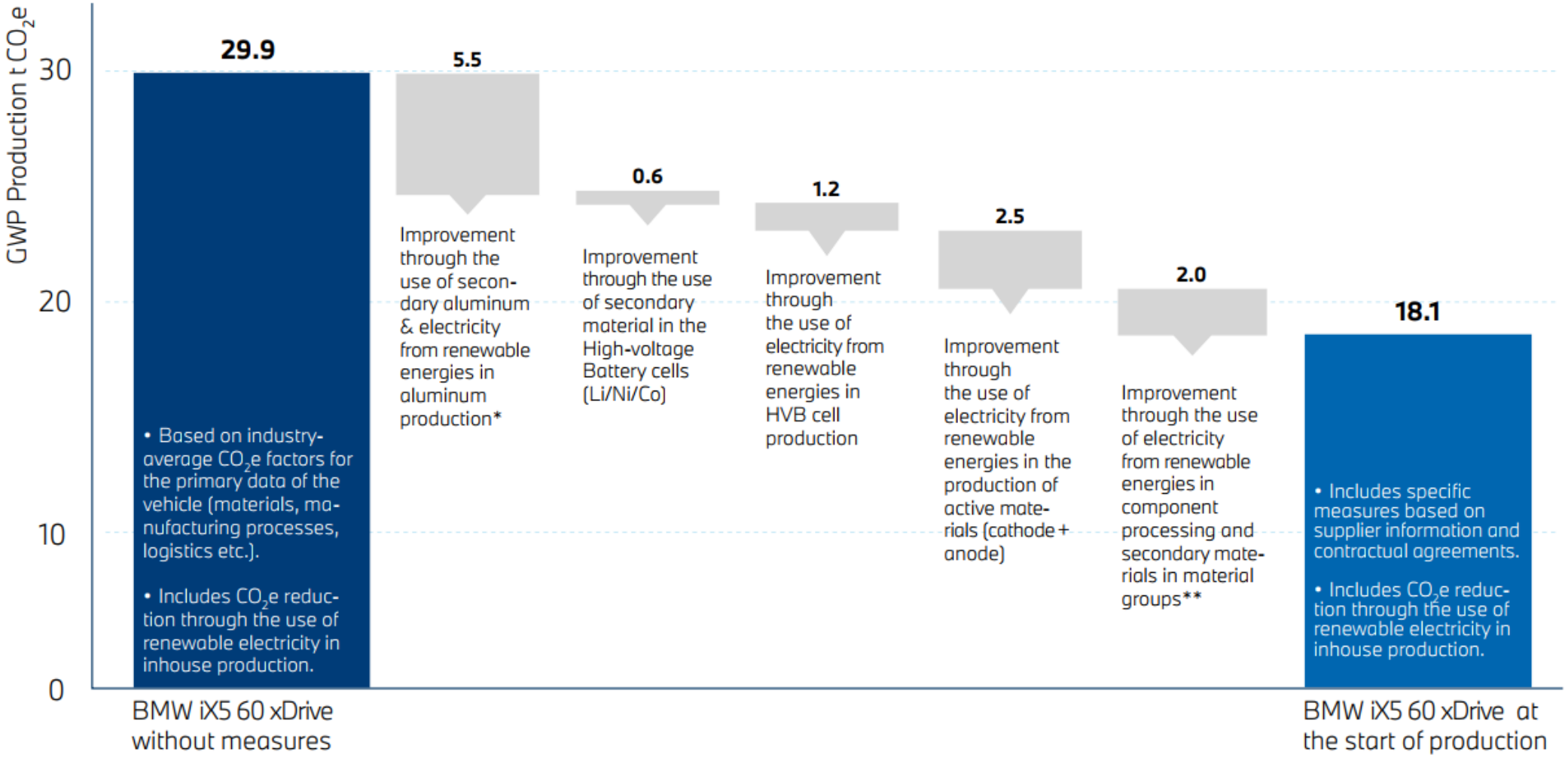


Fig. 5: Influence of development targets on the Carbon Footprint in the manufacturing phase of the BMW iX5 60 xDrive.
* Wheels, body and drivetrain components.
** Secondary material content in the following materials: steel, plastics, copper, glazing.

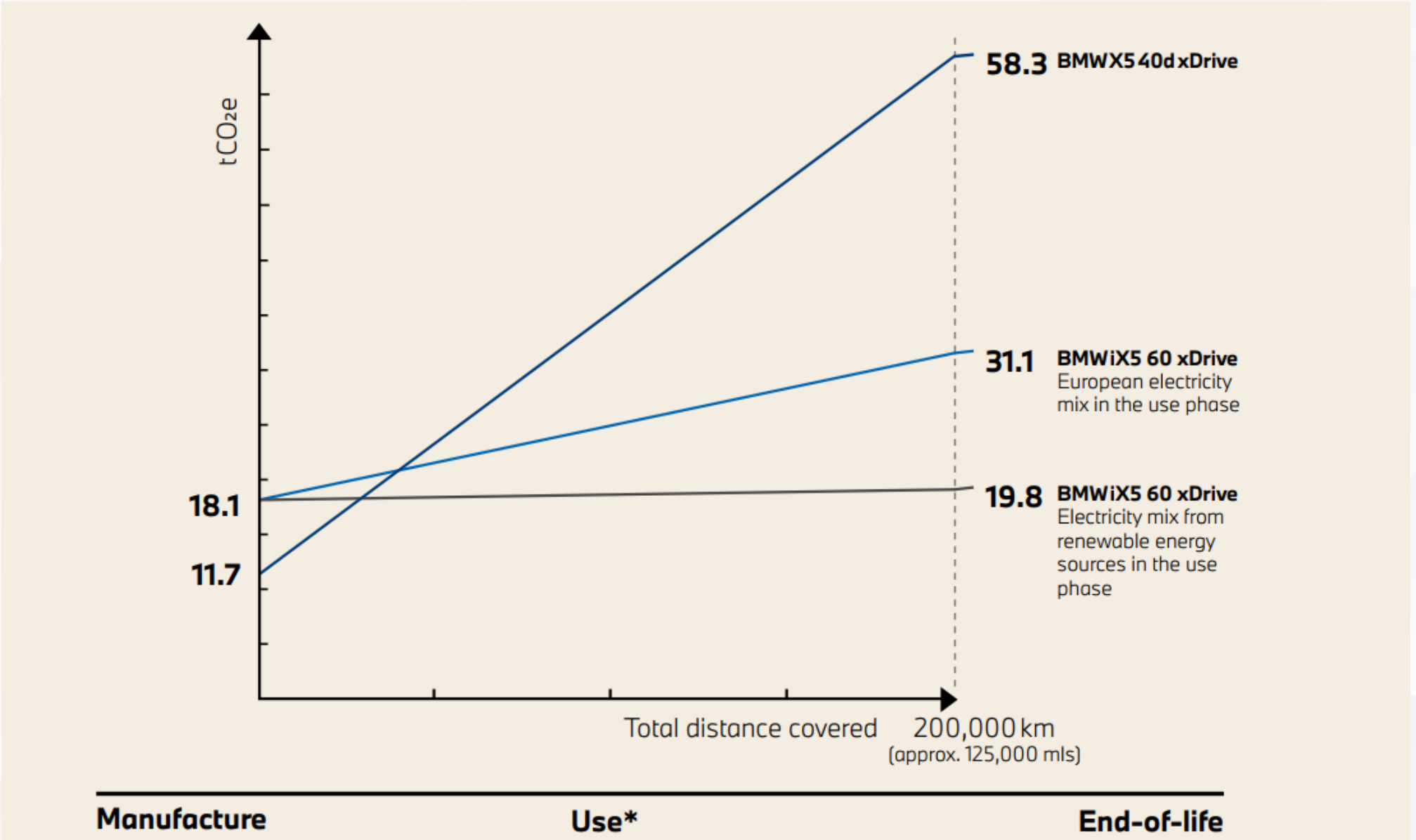


Fig. 4: Comparison of the Carbon Footprint of the BMW iX5 60 xDrive in relation to the BMW X5 40d xDrive calculated in accordance with IPCC AR6.
* Consumption data based on WLTP best-case value plus a segment-specific average mark-up for optional equipment in accordance with registrations in 2025 in EU27 (+UK, CH).

Q: In the automotive industry, regulators and many automakers are focusing on the transition from ICEs to EVs, which affects use-phase emissions. Why BMW is focusing on the emissions in the full value chain?

Exhibit 1 ICE Phase-out Announcements of Select Automotive Brands

Brand	By	Announcement
Bentley	2030	"Every car we create from 2030 onwards will be a battery electric vehicle."
Buick	2030	"The Buick brand is committed to an all-electric future by the end of this decade."
Ford	2030	"100% of passenger vehicles in Europe will be zero-emissions capable, all-electric or plug-in hybrid by 2026, and completely all-electric by 2030."
Mercedes-Benz	2030	"Mercedes-Benz will be ready to go all electric at the end of the decade, where market conditions allow."
MINI, Rolls-Royce	2030	"MINI and Rolls-Royce will be exclusively all-electric from the early 2030s onwards."
Volvo	2030	"Volvo Cars is committed to becoming a leader in the fast-growing premium electric car market and plans to become a fully electric car company by 2030."
Audi	2033	"As early as 2026, all new models from the Four Rings on the global market will be fully electric. And in 2033, the Four Rings will shut down the production of vehicles with internal combustion engines. One exception could be China, where the company is investigating the possibility of longer production depending on local demand."

Why is BMW focusing on emissions in the full value chain?

Financial Opportunities	Financial Risks
- Differentiated from other automaker (e.g., Tesla) and other BEV - Future customers will compare emissions in full value chain for BEV vs BEV	-Carbon tax/pricing - Higher emissions from Battery - Supply constraints and social risks associated with battery
Climate Impact	
- If not focus on full value chain, then carbon leakage - Can push supply chain to innovate - Potentially larger influence on supply chain	

Discussion: Should BMW Use PCF as Strategic Evidence?

- Should BMW make product carbon footprinting central to its EV strategy communication?
- Discuss from three angles:

1. **Technical credibility**

   Does PCF give a more complete picture than tailpipe emissions?

2. **Management usefulness**

   Does PCF help identify concrete decarbonization levers?

3. **Communication risk**

   Could PCF be challenged because of assumptions, boundaries, or data quality?

Why are we interested in LCA-based product carbon footprint?

- Carbon management perspective:
  - Which product should I buy (e.g. petrol car or electric car)?
  - What are the GHG emissions from my total consumption?
  - If I change the design of my product how will it affect the total emissions from the life cycle of the product?

- Carbon finance perspective:
  - Should I invest in this company/product (e.g. petrol car or electric car)?
  - Is the use-of-proceeds from this green bond really low carbon?
  - Will the product I'm investing in qualify for subsidies (e.g. under the Renewable Energy Directive)?

Wrap up

From Footprint to Action: PCF helps firms identify carbon hotspots across materials, production, use phase, and end-of-life, then prioritize decarbonization measures.

From Action to Evidence: Credible low-carbon product claims require transparent assumptions, supplier data, allocation rules, and verification.

BMW takeaway:

The key question is not simply whether BEVs are “cleaner,” but under what assumptions, through which life-cycle stages, and with which evidence BMW can reduce and prove product carbon performance.

Next Time – Policy/Project Level Carbon Accounting

- Essential reading:
  - Searchinger, T et al.. 2008. Use of U.S. Croplands for Biofuels Increases Greenhouse Gases Through Emissions from Land-Use Change. Science. 319 (5867), 1238–1240.
- Recommended reading
  - GHG Protocol Policy and Action Standard.
  - Brander, Matthew. 2017. Comparative analysis of attributional corporate greenhouse gas accounting, consequential life cycle assessment, and project/policy level accounting: A bioenergy case study. Journal of Cleaner Production. 167, 1401-1414
- Further reading:
  - ISO 14064-2: 2019 - Specification with guidance at the project level for quantification, monitoring and reporting of greenhouse gas emission reductions or removal enhancements





See you next time